Stellamarries Syombua

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Professional Summary

"Data Scientist, Biostatistician, and Healthcare Data Analyst with hands-on experience in statistical modelling, applied machine learning, and research analytics. I analyse large and complex datasets using Python, R, and SQL, translating data into insights that inform decisions across research and operational teams. My work focuses on practical data cleaning, exploratory analysis, and predictive modelling, with an emphasis on accuracy and interpretability. I'm comfortable presenting results to both technical and non-technical audiences and motivated by solving real-world research and business problems through data.

Experience

Yemaachi Biotech — Jul 2024 – Present
Data Analyst

- Performed end-to-end data analysis on breast and esophageal cancer projects, analysing patient demographics, risk factors, diagnostic stages, and treatment patterns to support research and decision-making.
- Developed and evaluated predictive models to identify trends and associations in cancer risk factors and clinical outcomes.
- Designed reproducible data cleaning and preprocessing workflows to ensure data quality, consistency, and analytical reliability.
- Conducted statistical analyses on electronic health record (EHR) data involving complex schemas and multi-source inputs.
- Developed analyses and models with an emphasis on reproducibility, interpretability, and downstream usability by other teams.
- Produced clear analytical reports and visualisations to communicate findings to both technical and non-technical stakeholders.
- Collaborated with multidisciplinary teams to define analytical questions and translate results into actionable insights.
- Utilised Python and R extensively for data manipulation, exploratory analysis, modelling, and visual analytics.
- Led statistical summaries of TACA study populations, analysing demographic, lifestyle, and clinical data for Ghana and Meru cohorts using `gtsummary`.
- Developed visual analytics comparing liver cancer (HCC) and hepatitis cohorts to support patient stratification.
- Co-developed an automated pipeline to harmonize multilingual (French and English) clinical datasets by standardizing categorical variables and reducing reliance on manual translation.

Adaptive Management and Research Consultants (AMREC) — May 2023 – Aug 2023
Research Intern

- Analysed quantitative and qualitative datasets using R, SPSS, Stata, and NVivo to support applied research projects.
- Supported research design, proposal development, and statistical reporting aligned with defined objectives.
- Worked both independently and within small teams to deliver high-quality analytical outputs under time constraints.
- Prepared reports and presented findings to stakeholders, ensuring clarity and relevance of insights.

Data Science Practice & Exposure

- Experience working across the full data science lifecycle, from problem definition and exploratory analysis to model development, evaluation, and communication of results.
- Built reproducible analytical workflows using well-structured, version-controlled code to support collaboration and reuse.
- Familiar with concepts around deploying, monitoring, and governing models in production environments through applied project work and continuous learning.
- Strong awareness of data quality, validation, documentation, and responsible use of models in real-world settings.

Personal Projects

- **Patient Risk Prediction System:** Built an end-to-end predictive pipeline classifying patient risk using structured clinical data, including data preprocessing, exploratory data analysis (EDA), feature preparation, model development, and evaluation to support risk stratification and clinical decision-making.
- **Lung Cancer Risk Analysis:** Analyzed symptom interrelations and behavioral correlates of lung cancer risk using correlation analysis in R to identify potential predictors.
- **Spatial Modeling of Diabetes in Kenya:** Applied spatial autocorrelation and clustering techniques (Moran's I) in R to identify geographic hotspots of diabetes prevalence.

- **Coffee Supply Chain Analytics:** Cleaned and analyzed farm-level data to optimize yield estimation, enumerator performance, and farming practices, supporting data-driven supply chain decisions.

Publications & Abstracts

- **Determinants of Triple-Negative Breast Cancer Among African Patients: Early Results from The African Cancer Atlas**. ASCO Annual Meeting Abstract; co-author.
  Link
- **Distribution and Clinical Correlates of Intrinsic Breast Cancer Subtypes in Meru County, Kenya**. Conference abstract; co-author.
- **Sociodemographic and Behavioral Factors Associated with Esophageal Cancer in Garissa County, Kenya**. Published article, Binaytara Journal.
  Link
- **FEMSAFE**: Sexual and Reproductive Health, STI, and Cervical Cancer Knowledge, Attitudes, and Practices Study. Manuscript under review at the African Population and Health Research Center (APHRC); co-author.

Education

Jomo Kenyatta University of Agriculture and Technology (JKUAT) Jul 2019 – Jul 2024
B.Sc. Biostatistics — Second Class Upper Division
Relevant Coursework: Regression Analysis, Survival Analysis, Epidemiology, Machine Learning, Clinical Trials

Citadel Maiden High School Jan 2016 – Nov 2019
Grade: B

Certifications

- Full Stack Development Internship Program — Edureka (2025)
- Power BI Workshop — Pragmatic Works (2024)
- Proposal Writing and Mobile Application Tools (CommCare and Kobo Collect) — AMREC
- Statistical Software (R, Stata, SPSS) — AMREC (2023)

Skills

- **Programming & Analysis:** Python (pandas, scikit-learn), R (tidyverse, survival, gtsummary), SQL
- **Analytics & Modelling:** Predictive Modelling, Regression, Survival Analysis, Statistical Inference
- **Data Science Workflow:** Data Cleaning, Feature Engineering, Exploratory Analysis, Reproducible Workflows, Documentation
- **Visualisation & Reporting:** Power BI, Excel
- **Research Tools:** Stata, SPSS, Kobo Toolbox, CommCare, NVivo
- **Languages:** English (fluent), Swahili (fluent)

Leadership & Activities

Candidate for Secretary General, JKUBSA (2023)

- Advocated for student welfare, policy reforms, and community engagement within the student body.

Volunteer, ABE Club

- Participated in initiatives supporting children with cancer, including outreach activities at Kenyatta National Hospital.

References

Available upon request.